

ASSIGNMENT #2

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
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Your name: Huda Usman
Due Date: Wednesday February 05 by 5:00pm in my office

Number of points.....100
Your score.....

ASSIGNMENT TOPIC

Review of Final Control Elements

Note: All problems are worth 100 points. Your final score will be the average value

Introduction

A final control element is a mechanical device that make changes in the process physically as a response to a variation in a process control system ^[2]. It is an instrument that adjusts the manipulated variable by taking a certain action to bring back the controlled variable to the set point. To achieve a desired output, the final control element is considered to be the most essential part of the process control system ^[1]. A few examples of the final control element include valves, servometers, and solenoids. However, this report will mainly focus on the types of valves used as a final control element. In a chemical plant, the valves can be broadly categorized in the following two main categories:

1. ***Shutoff valves:***

Shutoff valves are also known as the block valves or isolation valves. It is used to stop the flow and is expected to give a positive seal when closed and a very less resistance to flow when open ^[4]. The examples include gate, plug, and ball valves.

2. ***Control valves:***

Control valves are both manual and automatic, are used to regulate the flow in a process control scheme.

Types of Control Valves

The main types of valves and their industrial applications are:

1. ***Gate Valves:***

Like mentioned earlier, gate valves serve as an isolation valve which help in controlling the flow of the fluids through the pipes. These valves are secured into the pipeline to start or stop the flow of the fluids. Most commonly, they are used in commercial centers and our home pipelines. A few advantages of using the gate valve includes very low friction losses, minimum pressure loss, laminar flow, save energy and reduce total ownership cost, and it gives very low pressure drop when it is fully open. However, it does carry a few disadvantages such as, they cannot be closed and open quickly, they tend to cause vibrations, leakage can occur at high temperatures, maintaining and repairing of the gate valves is not straight forward ^[3]. In industries, gate valves are used in high pressure and temperature conditions.

2. ***Globe Valves:***

Globe valves are like the gate valves. It is a linear motion valve and it is designed to start, stop, and/or regulate the fluid flow in the pipes. It contains a moveable disk that can either be completely removed from the fluid flow or can close the path of the fluid flow. One of the main advantages of the globe valve is that it does not have a leakage problem as other valves does. It helps throttle the flow, it does not take much longer to open-close the globe valve, and it contains a positive shutoff ^[5]. However, the pressure drop is higher and under high pressures, it requires a tremendous amount of force. In industries, globe valves are used for throttling flow purposes.

3. **Plug Valve:**

Plug valve consists of tapered or a cylindrical plug that is used to regulate the fluid flow through the pipes. When open, the passage through the valve is aligned with the flow. The solid part of the plug is responsible to block the flow when the plug is turned 90° from the open position. Like globe valves, plug valves are also quick to close or open. It does not provide much resistance to flow, prevents leakage, and its maintenance is straight forward. However, due to higher friction, it requires higher force of operation, hence, greater pressure drop, it may be pricy as compared to the ball valves. In industries, plug valves can be utilized for high temperature and high pressure processes. Since it can provide a bubble tight shutoff, it is utilized in mud, slurries and other sewage service applications [6].

4. **Ball Valve:**

This valve uses a ball to regulate the flow of the fluid (liquid/gas). They provide better sealing by keeping the hole either closed, open or partially open therefore, is a good choice to use with gases. They are very easy to maintain, operate and repair. They can support the pressure and temperature up to 700 bars and 200 °C respectively [5]. However, they cannot be utilized for permanent throttling, and with the residual fluids that can cause leakage problems. They are used in NG (natural gas) and oil industries to maintain the pressure and the flow for mud, slurries, and other gases and liquids [5].

5. **Diaphragm Valve:**

Diaphragm valve are generally two-directional as opposed to no return valve. They can be utilized for throttling and stop-start applications. They can be used when dealing with low pressure systems. They provide a smooth laminar flow that decreases the pressure drop [8]. They are designed such that they are completely leak proof and therefore, is used for the processes where the leakage tolerance is very low. Hence, in industries, they are utilized in food, pharmaceuticals, brewing and other systems to avoid contamination [8]. However, these valves have a restriction to be used at certain pressures and temperatures depending on the material of construction of the diaphragm. Also, they are not built in wide range of sizes [5].

6. **Butterfly Valve:**

Butterfly valves have a limited control capability to regulate the flow. They are quite accurate and reliable to be used for industrial applications. They can throttle the flow across the pipes. We can remove or install butterfly valves without dislocating the whole pipeline scheme. However, they do not have tight shutoffs. They can be utilized in pharmaceuticals, food processing and chemical industries [5].

7. **No Return Valve (NRV):**

NRVs are also known as check valves and are one-directional valve that allow fluids to only flow through in a single direction. They prevent backward flow through the pipes which can damage the equipment [5,9]. Their main application is protecting pumps and compressors from shutting down if backward flow take place. They help the system sustain the pressure and serve as a backup plan for the process control [9]. However, the closing element of the NRVs can crash and cause damage to the system.

Final Control Elements Control Loops

For feedback control, the controller computes the error between input and setpoint and based on a response, adjusts the output using the defined algorithm. Figure 1.1 illustrates the feedback control mechanism.

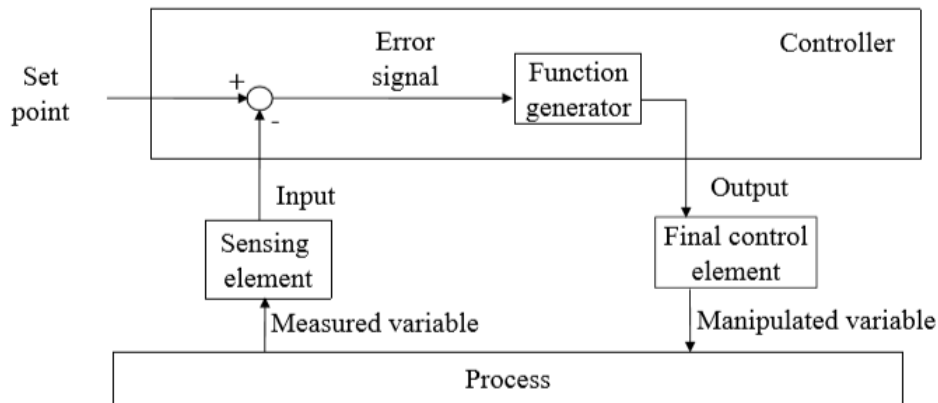


Figure 1.1: Feedback Control Loop ^[4].

For feedforward control, the control system senses the disturbance and adjusts the manipulated variable to compensate for that disturbance, so it does not affect the control output. Figure 1.2 illustrates the feedforward mechanism ^[7].

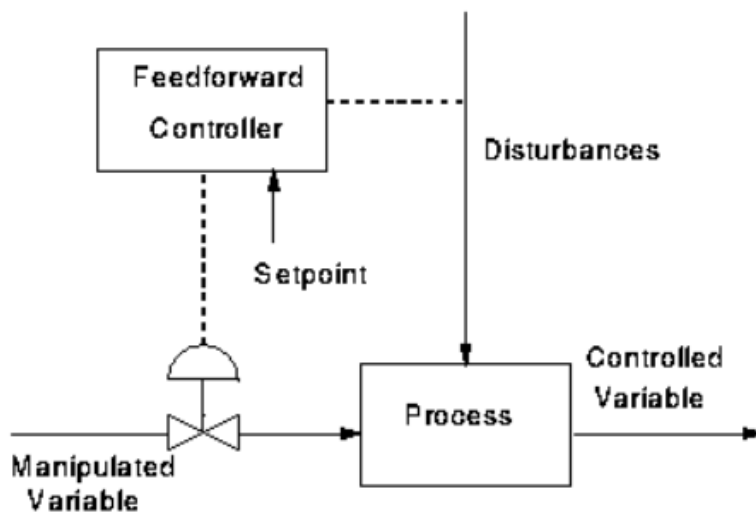


Figure 1.2: Feedforward Control Loop ^[4].

References

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- [9] “BM Engineering Explain the Functions and Advantages of Non-Return Valves.” *Process Industry Forum*, www.processindustryforum.com/article/bm-engineering-explain-the-functions-and-advantages-of-non-return-valves.